PranaTwin-T2D

A clinician-facing research sandbox for short-horizon glycemic forecasts

Historical context.
An evolving virtual state.
A forecast that can be audited.

Happiest Health Digital Twin Challenge 2026
Team / institution: [FILL IN] | Research only

The focused use case

Forecast glucose trajectories up to 120 minutes in a bounded software demonstrator

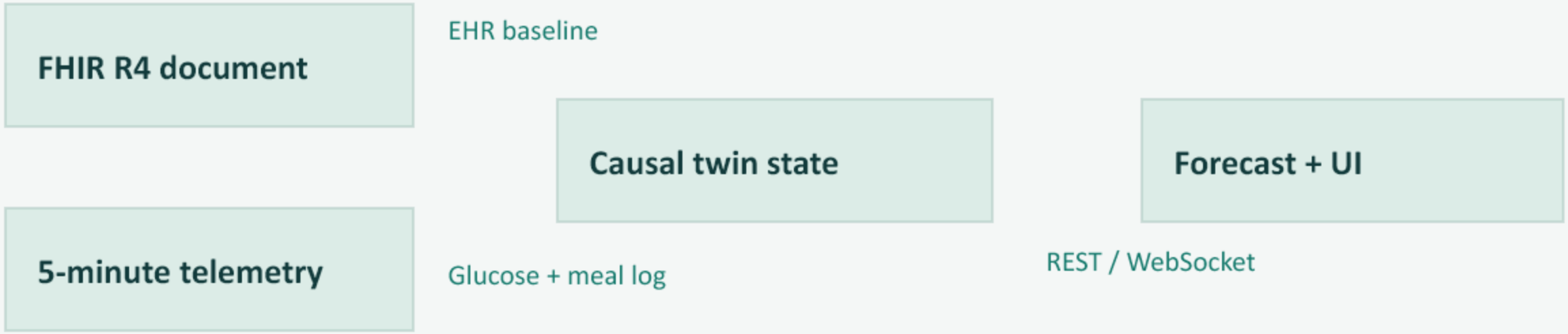
01 Research flags: glucose above 180 or below 70 mg/dL.

02 Three fictional outpatient contexts reflect comorbidity, waist/BMI and shift-work fields.

03 India-specific context does not imply ethnicity-based physiology or a representative cohort.

Two streams update one state

The historical glucose Observation initializes the state used by the replay



One FastAPI process serves the React UI. Consent gates patient routes and streams.

A transparent reduced-order model

Illustrative engineering dynamics with explicit units and bounds

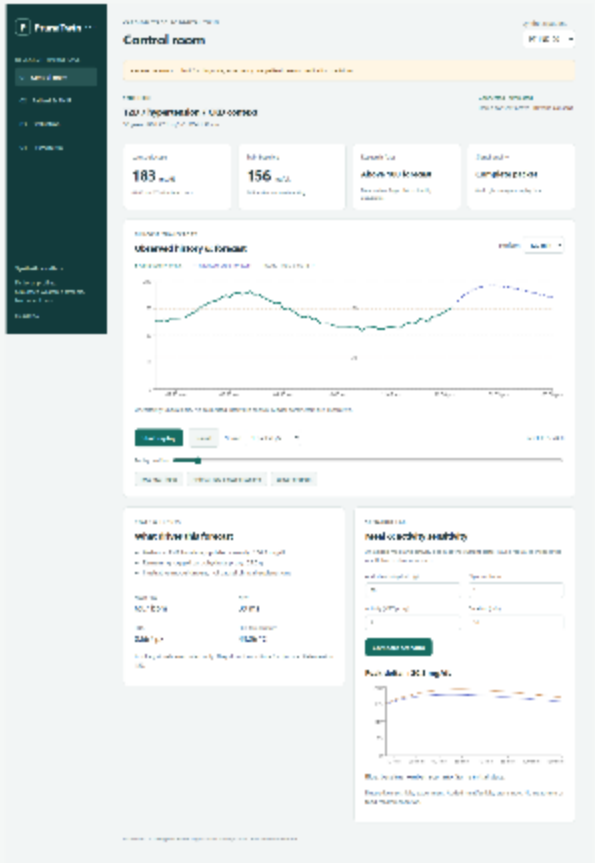
01 Logged carbohydrate decays through a first-order proxy compartment.

02 Glucose relaxes toward the EHR-derived, causally updated baseline.

03 Each observation updates the state. No insulin or drug-response model is claimed.

The working control room

Replay, inspect and compare from a consistent synthetic patient state



Start / pause / speed / seek

FHIR inspection

Bounded meal + activity scenarios

Visible source and uncertainty limits

Evaluation prevents future leakage

Within-patient chronological splits with a two-hour purge at each boundary

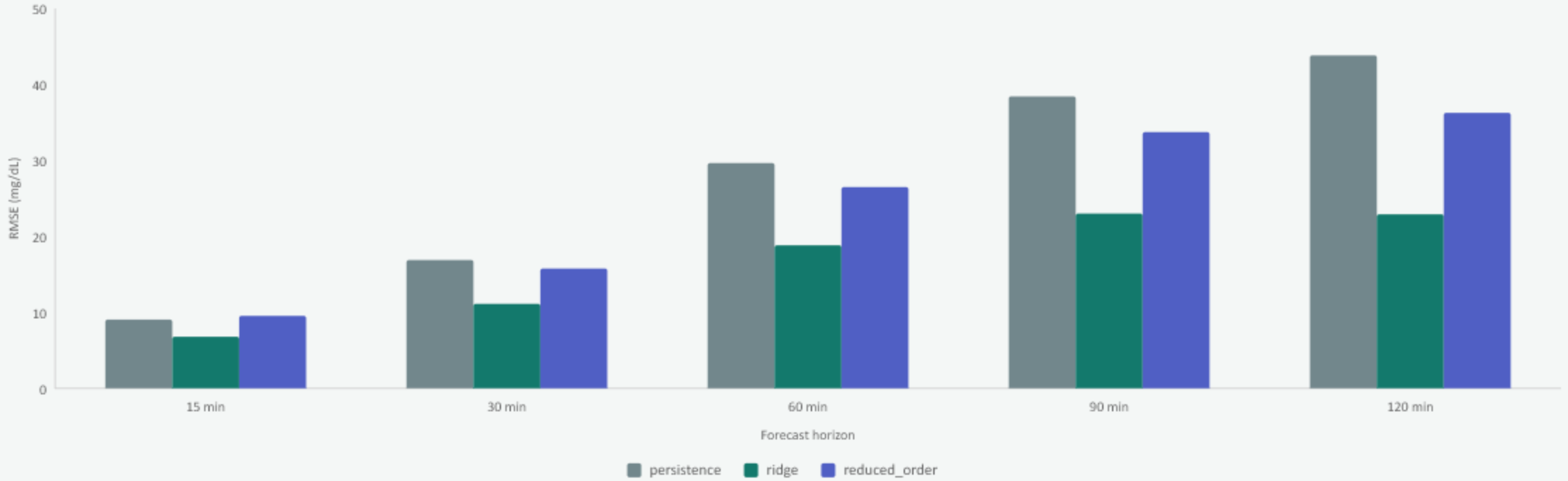
01 Training 60%, reserved validation 20%, final test 20%.

02 Fit ridge preprocessing and regression on training rows only.

03 Compare identical held-out targets at 15, 30, 60, 90 and 120 minutes.

Regression is strongest on this simulator

RMSE in mg/dL; 276 held-out predictions per model and horizon



Synthetic results cannot establish clinical validity. Reduced-order also loses to persistence at 15 minutes.

Uncertainty and failure modes

The interface suppresses unsupported confidence bands and risk percentages

01 No calibrated intervals, probabilistic Brier score or verified Clarke grid.

02 Unannounced meals and the artificial low scenario expose forecast limitations.

03 Missing, stale and unsupported inputs trigger explicit quality states or errors.

Prototype governance

Revocation blocks protected endpoints and closes active live subscriptions

01 FHIR viewer reports local structural checks only. No full ABDM conformance claim.

02 Consent status and expiry are checked server-side. Demo restoration is intentionally accessible.

03 No identity system or compliance certification. Keep the sandbox on a trusted local computer.

What remains before submission and clinical research

Working software is separate from evidence of clinical performance

01 Complete team details, publish the repository and record the required 20-minute minimum video.

02 Validate on authorized real CGM data with a prespecified subject holdout.

03 Integrate full FHIR validation, identified physiology and independent security review before clinical translation.